

PERVEZ ALI

Ph.D. Candidate in Mathematics

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Google Scholar 

RESEARCH INTERESTS

Design and analysis of Langevin-type Markov chain Monte Carlo algorithms, especially nonreversible dynamics and sampling on constrained domains; nonasymptotic convergence guarantees and large deviations analysis of samplers, paired with careful numerical studies; stochastic optimization and sampling for Bayesian inference and generative modeling.

EDUCATION

Florida State University

Ph.D. in Mathematics

Expected June 2027

Tallahassee, FL, USA

- Advisor: Lingjiong Zhu
- Dissertation: *Non-Reversible Langevin Dynamics for Unconstrained and Constrained Sampling*
- Fulbright Scholar

Sukkur IBA University

M.S. in Mathematics

2019

Sukkur, Pakistan

- Focus: probability, stochastic processes, and mathematical finance (option pricing)

University of Sindh

B.S. in Mathematics

2016

Jamshoro, Pakistan

PUBLICATIONS AND PREPRINTS

- [1] **Pervez Ali**, Xiaoyu Wang, Yingli Wang, and Lingjiong Zhu. *Accelerating non-smooth and heavy-tailed sampling*. Preprint, [arXiv:2610.11139](#), 2026.
- [2] **Pervez Ali**, Weihao Dong, and Xiaoyu Wang. *Penalized nonreversible Langevin for constrained sampling*. Preprint, [arXiv:2609.25381](#), 2026.
- [3] Nian Yao, **Pervez Ali**, Xihua Tao, and Lingjiong Zhu. *Accelerating Langevin Monte Carlo sampling: A large deviations analysis*. Preprint, [arXiv:2503.19066](#), 2025.
- [4] Javed Hussain and **Pervez Ali**. *On modelling and pricing weather derivatives driven by nonlinear Brownian motion*. *International Journal of Analysis and Applications* **19**(1):29–46, 2021.

TALKS

- *Accelerating Langevin Monte Carlo sampling: A large deviations analysis*. May 2026
5th International Conference on Computing, Mathematics and Engineering Technologies (iCoMET 2026), Sukkur IBA University
- *On modelling and pricing weather derivatives driven by nonlinear Brownian motion*. Jan 2020
iCoMET 2020, Sukkur IBA University

CONFERENCES AND SCHOOLS ATTENDED

- Math in Industry Symposium, Florida State University Feb 2025, Feb 2026
- Second Joint Alabama–Florida Conference on Differential Equations, Dynamical Systems and Applications, Florida State University May 2024
- Southeast Applied and Computational Math Student Workshop, Georgia Tech, Atlanta, GA Apr 2024
- AMS Spring Southeastern Sectional Meeting, Florida State University Mar 2024

- Hausdorff School: Formal Mathematics and Computer-Assisted Proving, University of Bonn Sep 2023
- Florida–Georgia Applied Computational Math Student Workshop, Florida State University Apr 2023
- Winter School on Risk, Stochastic, Finance and Insurance, Sukkur IBA University Dec 2018
- iCoMET 2018, Sukkur IBA University Mar 2018

TEACHING EXPERIENCE

Florida State University Aug 2022 – present
Graduate Teaching Assistant and Instructor Tallahassee, FL, USA

- **Instructor:** Calculus I (Spring 2025, Summer 2025), Calculus II (Fall 2025, Spring 2026), Calculus III (Summer 2026, Fall 2026).
- **Teaching assistant:** Calculus, Trigonometry, College Algebra; tutor, Math Resource Center.

Sukkur IBA University Aug 2017 – Jul 2022
Lecturer in Mathematics Sukkur, Pakistan

- **Mathematics:** Calculus I–III, Linear Algebra, Ordinary Differential Equations, Scientific Computing.
- **Probability and statistics:** Probability, Statistics, Inferential Statistics.
- **Business and actuarial:** Business Mathematics, Business Statistics, Actuarial Mathematics.

FELLOWSHIPS AND AWARDS

- Educator of the Year Award 2023
- Travel grant to attend the Hausdorff School, University of Bonn 2023
- Fulbright Scholarship for Ph.D. study in the United States 2022
- Departmental Merit Scholarship, Sukkur IBA University 2019
- Prime Minister’s Scholarship for M.S. study (fully funded) 2018
- Student of the Year in Mathematics, University of Sindh (awarded a laptop) 2015

SERVICE, OUTREACH, AND MENTORING

- Vice President, SIAM Student Chapter, Florida State University 2023 – 2024
- Fulbright outreach: led seven information sessions on the Fulbright program at three institutions.
- Mentored undergraduate projects at Florida State University and Sukkur IBA University.

COMPUTING AND LANGUAGES

Computing: Python (NumPy, SciPy, PyTorch, Matplotlib), MATLAB, $\text{\LaTeX}$.

Languages: English (fluent), Urdu (native), Sindhi (native), Punjabi.